

Saber Prototype

Why being fast is not good enough to steal a base

Sprint speed is the most intuitive base-stealing metric and the least essential one. On 10,844 tracked attempts, the ground a runner wins between the pitcher's first move and his release is worth **3.19×** the odds of being safe per standard deviation. Sprint speed is worth **1.40×**. Speed is real, and it is roughly 2.3 times less important than what the runner does once the pitcher commits.

What speed does buy is the green light. Across 551,823 runner-on-first opportunities, the fastest fifth of the league attempts a steal **7.1×** as often as the slowest fifth — on a near-identical share of chances. **Speed earns the attempt; technique converts it.**

1 · The same production, from opposite sources

Net stolen bases hide this completely. Two runners can finish four bases apart and be doing entirely different things. The model splits every runner's production into what his raw speed alone buys (NetSpeed) and what his skill adds on top (Steal+); the two sum to his net bases exactly.

Runner	Season	Sprint	Net bases	From speed	From skill	Skill share
Elly De La Cruz	2024	30.0 ft/s	51	52.1	-1.1	-2%
Shohei Ohtani	2024	28.1 ft/s	55	36.8	+18.2	+33%
Josh Naylor	2025	24.4 ft/s	28	16.0	+12.0	+43%

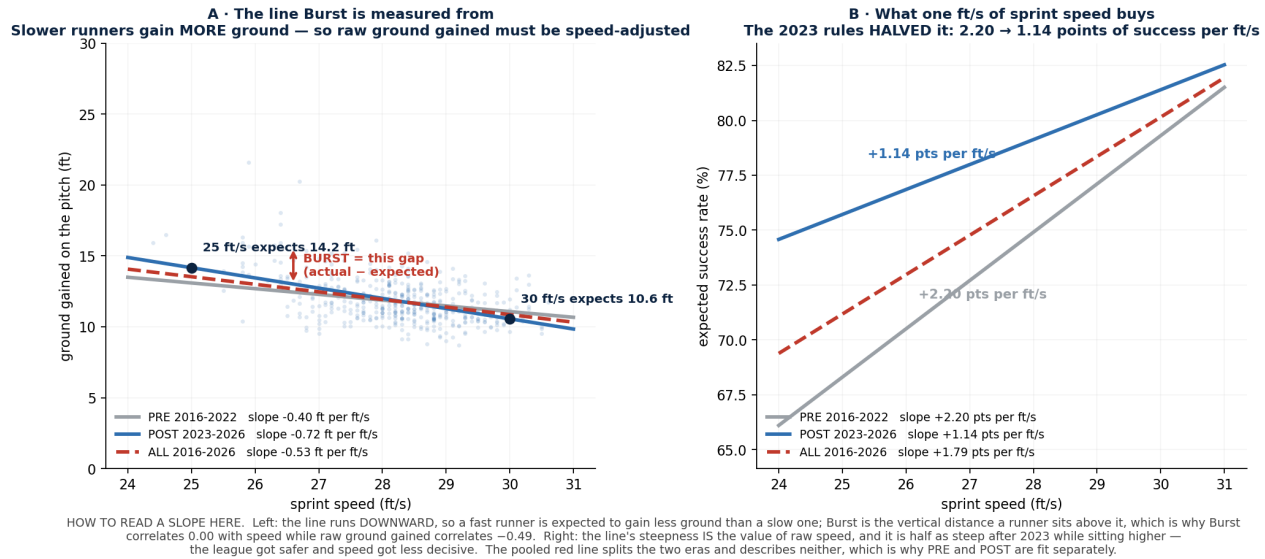
Elly De La Cruz and Shohei Ohtani finished 4 bases apart in the same season. De La Cruz's 51 are 102% wheels — his Steal+ is -1.1, meaning he produced marginally less than his speed alone predicts. Ohtani's 55 carry +18.2 bases of skill, the highest in the pool. They should not be projected forward the same way, and a stolen-base column cannot tell them apart.

2 · The runner the metric was built for

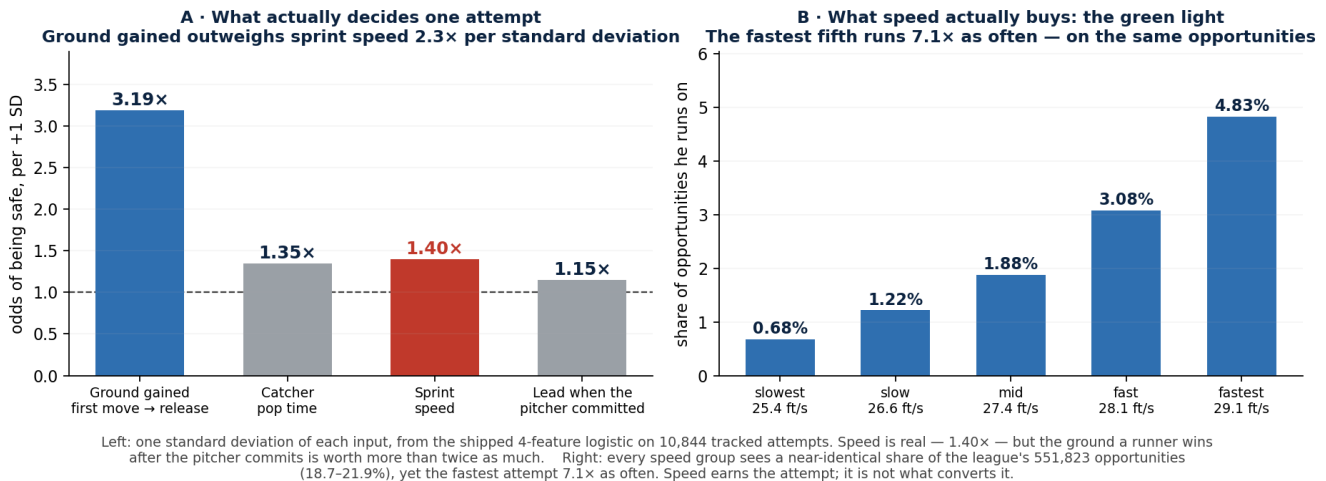
Josh Naylor ran **24.4 ft/s** in 2025 — slower than **98%** of the 579 runners Statcast timed that year — and went **30 for 32**. His Steal+ of **+12.0** is the **99th percentile**. He wins 15.9 feet before the ball leaves the hand, the most of anyone in the pool. On a raw speed leaderboard he is invisible.

3 · A natural experiment: the 2023 rules cut the value of speed in half

The disengagement limit, larger bases and the pitch clock changed what wheels are worth. Fitting the same model on each side of the break: one extra ft/s of sprint speed bought **2.20** percentage points of success before 2023 and only **1.14** after — a **48% reduction** — while the league's floor rose from 76.4% to 80.4%. When the environment hands everyone a margin, already having one is worth less. The premise of this project got stronger, not weaker.



4 · Speed decides whether he goes, not whether he makes it

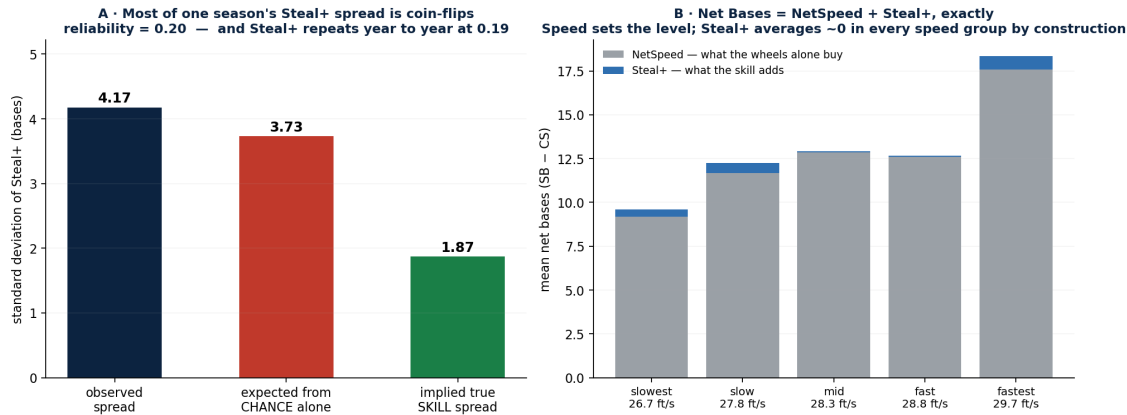


5 · What is actually coachable, and what actually repeats

A metric worth coaching has to be separable from the body it sits in and it has to persist. Burst — the ground a runner gains above what his own sprint speed predicts — is both, and it is the most repeatable number in the project.

	Correlation with sprint speed	Repeats year to year
Net bases (SB – CS)	+0.29	+0.38
Steal+ (production above speed)	-0.01	+0.19
Burst (technique above speed)	-0.00	+0.46

Steal+ describes the season it is computed on extremely well — it correlates +0.89 with success rate and +0.64 with net steals — while carrying essentially no speed signal. Burst barely describes outcomes at all, and is the thing that comes back next season.



Left: under the null that a runner is exactly as good as his speed predicts, SB ~ Binomial(attempts, p), so chance alone produces an SD of 3.73 bases at these volumes. Only 20% of the observed variance survives as skill — which is why Steal+ describes a season well ($r = 0.64$ with net steals) but forecasts the next one weakly. Right: the two components sum to net bases with zero residual ($< 1e-9$).

Right panel: net bases climb steeply with speed and NetSpeed tracks that climb almost exactly. Steal+ averages near zero in every speed group — by construction, which is what makes it a fair comparison across bodies.

6 · What this does not show

Three limits, stated plainly, because each one is the first thing a good analyst will ask about.

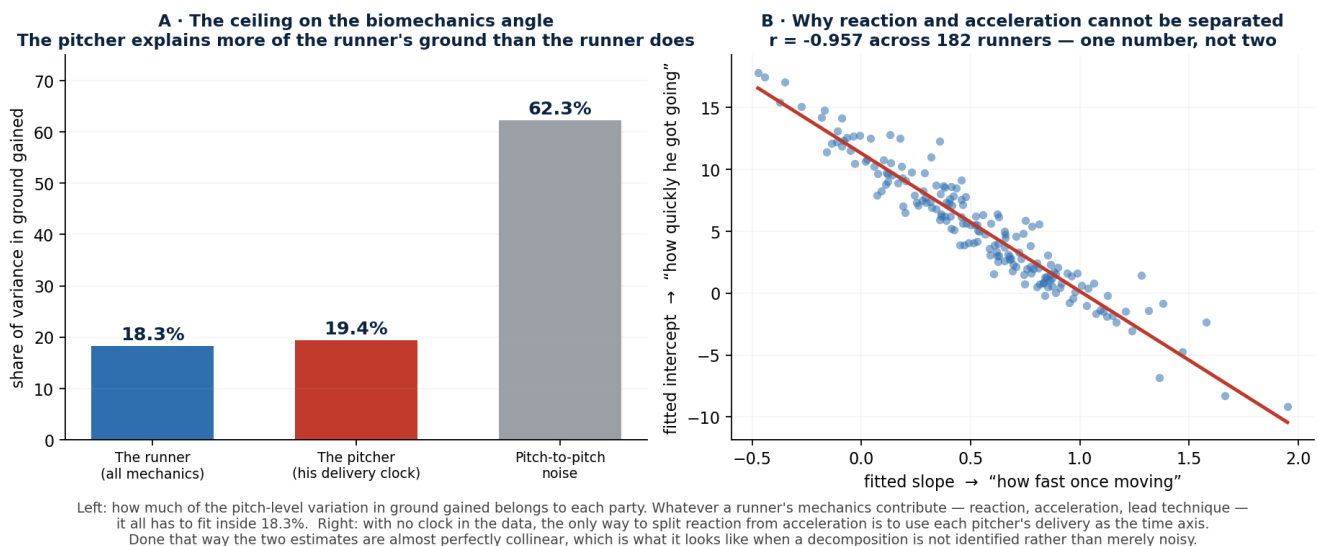
A single season of Steal+ is mostly luck. Steal+ is built from roughly seventeen binary outcomes at the median. Under the null that a runner is exactly as good as his speed predicts, chance alone produces a spread of 3.73 bases against an observed 4.17 — so only **20%** of the variance is skill. The reassurance is that this noise model, computed from attempt counts alone, independently reproduces the observed year-to-year correlation (0.201 predicted vs 0.192 measured). Only 7.1% of runner-seasons separate from average by more than two chance standard deviations. Name those; do not rank the rest.

The per-attempt model describes attempts that were chosen. Its inputs only exist once a runner has already gone, and §4 shows selection is heavy — a 7.1× gradient by speed. The odds ratios describe attempts that survived that filter, not hypothetical ones.

The 2023 comparison is observational. One discontinuity, with rosters and organisational aggressiveness changing across it too. The direction and rough magnitude are credible; “halved” is not a causal estimate.

7 · The measurement this project cannot make

The natural next question is whether the ground a runner wins is *reaction* to the pitcher's cue or *acceleration* once moving. That decomposition is not identified here, and the reason is structural rather than a matter of effort.



7 · continued — why there is no clock

Ground gained is a **distance over an interval whose duration is never recorded**. That interval is the pitcher's own first-move-to-release time, which varies by pitcher, by pitch, and by whether he slide-steps. Statcast's per-attempt feed carries no timestamp for it. The candidate substitutes all fail for stated reasons:

Candidate clock	Why it does not work
Savant pitch tempo	The endpoint ignores its year parameter — it returns one season's data for every season requested.
“Jump time” in the running splits	It is a batted-ball home-to-first split, not a reaction to a pitcher. It correlates -0.61 with sprint speed.
Release extension / speed	Both describe ball flight after release — the wrong side of the interval.
Each pitcher's delivery as the axis	Tried. Fitting a per-runner slope and intercept against a leave-one-out pitcher clock gives $r = -0.957$ between them across 182 runners — one number, not two.

There is also a ceiling worth knowing before investing in mechanics at all. Of the pitch-level variation in ground gained, **18.3%** belongs to the runner and **19.4%** to the pitcher, with the rest pitch-to-pitch noise. Whatever reaction, acceleration and lead technique jointly contribute, it has to fit inside that 18%.

Consistent with that, the acceleration-flavoured variables that *are* in the data show no relationship with Burst across 408 runner-seasons: accel-gap -0.072 , jump time $+0.073$, acceleration phase $+0.087$. They are all strongly tied to sprint speed instead (-0.70 for the acceleration phase). **The lab variables available here measure how fast a runner is, not how well he starts.**

That is the case for the instrument, not against the question. A reaction latency to a genuine reactive cue is exactly what a lab protocol can measure and a broadcast tracking feed structurally cannot. A dirt-surface trial with a reactive start is motivated by a measured gap rather than a hunch — and it is the only way to find out whether the acceleration mechanics matter here at all.

Speaker notes

Open on three numbers.

Number	What it says
3.19× vs 1.40×	Ground gained versus sprint speed, per standard deviation, on the same attempts.
-1.1 vs +18.2	De La Cruz and Ohtani, 51 and 55 net bases in the same season — one is all wheels, one is a third technique.
2.20 → 1.14	Points of success per ft/s, before and after 2023. The rules cut the value of speed roughly in half.

Expect two objections. Both have answers.

“Isn’t a single season of this mostly noise?” — Yes, and that is measured rather than hand-waved: reliability 0.20. The reason to trust it is that a binomial noise model built from attempt counts alone predicts the year-to-year correlation that is actually observed, to within 0.009. So quote Steal+ with its error bar, and lean on Burst — +0.46 year to year — when the question is about a runner’s process.

“You can’t actually measure the mechanics.” — Correct, and concede it immediately. Then use it: the ceiling is known, the clock does not exist in this data, and that is precisely the argument for collecting it properly. Page 5 is the ask.

The one-line version.

Speed earns the attempt. Technique converts it.

Sources — all figures and values regenerate from the model repository: 10,844 tracked attempts and 452 qualified runner-seasons (2023–2026), 551,823 opportunities for the selection analysis, and 690+452 runner-seasons for the era comparison. Full specification, robustness and limitations in the v15 report.